EMA51005 – Discrete Mathematics (Question Bank)

UNIT-II Combinatorics
Part – A

1. Assuming that repetitions are not permitted, how many four-digit numbers can be formed from the six digits 1, 2, 3, 5, 7, 8?
2. Show that if seven colours are used to paint 50 bicycles, at least 8 bicycles will be the same colour.
3. How many 7 letter words can be formed using the letters of the word BENZENE?
4. State Pigeonhole principle.
5. Find the minimum number of students need to guarantee that five of them belongs to the same subject having major as English, Maths, Physics, and Chemistry. .
6. Define the generalisation of the pigeonhole principle.
7. A man hiked for 10 hours and covered a total distance of 45 km. It is known that he hiked 6 km in the first hour and only 3 km in the last hour. Show that he must have hiked at least 9 km within a certain period of 2 consecutive hours.
8. Define Recurrence relation.
9. Write the general solution of recurrence relation.
10. Write the principle of Inclusion and Exclusion formula for the term $|A \cup B \cup C|$.

Part – B

1. Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3, for $n \geq 1$.
2. Prove by mathematical induction that $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$.
3. A club consisting of 6 men and 7 women, in how many ways can we select a committee of 4 persons which has at least one woman?
4. There are 250 students in an engineering college. Of these 188 have taken a course in Fortran, 100 have taken a course in C and 35 have taken courses in Java. Further 88 have taken courses in both Fortran and C. 23 have taken course in both C and Java and 29 have taken courses in both Fortran and Java. If 19 of these students have taken all the three courses, how many of these 250 students have not taken a course in any of these three programming languages?
5. Suppose there are 6 boys and 5 girls.

   (i) In how many ways can they sit in a row.

   (ii) In how many ways can they sit in row if the boys and girls each sit together.

   (iii) In how many ways they can sit in a row if the girls are to sit together and the boys do not sit together.
6. Solve the recurrence relation $a_{n+2} - 6a_{n+1} + 9a_n = 3X2^n + 7X3^n, n \geq 0$, given that $a_0 = 1 \; and \; a_1 = 4$.
7. How many positive integers n can be formed using the digits 3,4,4,5,5,6,7, if n has to exceed 50,00,000?

HITS MATHEMATICS

8. If 6 people A,B,C,D,E,F are seated about a round table, how many different circular arrangements are possible, if arrangements are considered the same when one can be obtained from the other by rotation ? If A,B,C are females and the other are males, in how many arrangements do the genders alternate ?
9. A team of 11 players is to be chosen from 15 members. In how many ways can this be done if
   i) One particular player is always included.
   ii) Two such players have always to be included.
10. Solve the recurrence relation $a_n - 2a_{n-1} = 3^n, a_1 = 5.$

Part – C

1. Solve the recurrence relation $a_{n+1} - a_n = 3n^2 - n; n \geq 0, a_0 = 3.$
2. Use mathematical induction to prove that
   $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9, for $n \geq 1$.
3. Find a formula for the general term a_n of the Fibonacci sequence 0,1,1,2,3,5,8, 13,.
4. Prove by mathematical induction that $1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$.
5. Use mathematical induction to show that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$, for $n \geq 2$.
6. Show that $a^n - b^n$ is divisible by $(a-b)$ for all $n, n \geq 1$ using principle of mathematical induction.
7. Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2, 3, 5 and 7.
8. How many bit strings of length 10 contain
   (a) Exactly four 1's
   (b) Atmost four 1's
   (c) At least four 1's
   (d) An equal number of 0's and 1's ?
9. Solve the recurrence relation $a_n = 2(a_{n-1} - a_{n-2}), n \geq 2$ and $a_0 = 1$, $a_1 = 2.$
10. In how many ways can 20 students out of a class of 30 be selected for an extra -curricular activity, if
    (a). Rama refuses to be selected ?
    (b). Raja insists on being selected ?
    (c). Gopal and Govind insist on being selected ?
    (d). Either Gopal or Govind or both get selected ?

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